

ABSTRACT

This study develops a machine learning-based multi-sensor Drought Index (DI) approach for agricultural drought assessment in the Gaya district of South Bihar, a region with no canal irrigation and strong dependence on rainfall. Multi-source satellite data, including SAR backscatter, vegetation and moisture indices, land surface temperature, precipitation, soil moisture, and atmospheric parameters, were integrated within a single modeling approach to represent different components of drought. These remote sensing datasets were used to train and evaluate multiple machine learning models, including Decision Tree, Random Forest, XGBoost, Multi-Layer Perceptron, and LightGBM, to estimate the Normalized Difference Moisture Index (NDMI) as a measure of soil moisture conditions. Among the models, LightGBM performed best, achieving an R^2 of 0.97 during training and 0.89 during testing, with corresponding RMSE values of 0.0217 and 0.0415, indicating strong predictive performance. Feature Importance (FI) derived from the model was used to quantify the contribution of each variable and to construct a multi-sensor Drought Index (DI). The proposed DI replaces fixed weighting with a data-driven approach to capture non-linear relationships among multiple environmental variables. The resulting DI effectively captured spatial differences in drought conditions and identified areas with higher drought severity, consistent with variations in vegetation condition and moisture patterns. The proposed method provides a reliable and scalable approach for agricultural drought assessment in rain-fed and data-scarce regions.

1.Introduction

Agricultural drought has emerged as a major global crisis, affecting nearly 45% of cultivated land and significantly disrupting food production systems [34]. Since 1980, over 37% of the Earth's land has experienced notable soil moisture loss, while drought events have increased by 29% since 2000. These conditions impact more than 55 million people annually and can reduce crop yields by up to 22% in dry years, threatening the livelihoods of approximately 1.3 billion people. The burden is especially severe in developing countries, where agriculture accounts for 82% of drought-related losses [33]. Driven by climate change and human activities, including altered precipitation, rising temperatures, and groundwater overuse, drought impacts are intensifying through reinforcing climate-agriculture feedbacks. Projections indicate that by 2050, up to three-quarters of the global population may be affected [33], highlighting agricultural drought as a critical threat to food security and ecological stability.

The impact of agricultural drought is particularly severe in rain-fed systems, where crop production depends strongly on rainfall timing and distribution [8]. Even short-term variations during sensitive growth stages can result in significant soil moisture deficits and yield loss [5]. This vulnerability is further increased by land degradation and inefficient water management practices, which reduce the resilience of agricultural systems [21]. Because of this, there is a clear need for monitoring approaches that can capture both spatial variability and temporal changes in drought conditions [9].

Traditional drought assessment methods mainly rely on ground observations and meteorological indices such as the Standardized Precipitation Index (SPI) and Palmer Drought Severity Index (PDSI) [4]. While these methods are widely used, they are limited by sparse station networks and uneven data coverage [17]. More importantly, they do not directly represent soil moisture availability or crop water stress, which are essential for agricultural drought assessment [26]. As a result, they are often not suitable for field-level analysis, especially in regions with high spatial variability [9].

Remote sensing provides a more practical alternative by enabling continuous and spatially consistent observation of environmental variables [13]. Satellite data allow monitoring of vegetation condition, land surface temperature, precipitation, and soil moisture across different spatial and temporal scales [2]. Vegetation indices such as NDVI are widely used because they respond to changes in plant condition [12]. Combined drought indices that integrate vegetation, temperature, and soil moisture have shown better performance than single-variable approaches [10]. However, many of these approaches rely on fixed relationships, which limits their ability to represent complex and non-linear environmental interactions [14]. This limitation becomes more significant at finer spatial scales, where variability in soil properties and crop conditions strongly affects drought response [9].

These challenges are clearly observed in semi-arid regions such as the Gaya district in Bihar, India, which covers approximately 4,976 km² and is dominated by rain-fed agriculture. The absence of canal-based irrigation systems makes crop production highly dependent on rainfall, increasing sensitivity to soil moisture variability. Under such conditions, even short-term climatic fluctuations can lead to severe drought impacts. Existing drought monitoring approaches are not able to capture these fine-scale variations effectively, which highlights the need for improved methods [11].

1.1 Objectives of the Study

Based on these limitations, this study aims to develop a data-driven multi-sensor remote sensing framework for agricultural drought assessment.

The specific objectives are:

- To integrate multi-sensor datasets, including vegetation indices, soil moisture, land surface temperature, and precipitation, for comprehensive drought characterization.
- To develop a machine learning-based model capable of capturing non-linear relationships among drought-related variables.
- To improve the spatial resolution and predictive accuracy of drought detection at finer scales.

To develop a scalable and robust framework applicable across different agro-climatic regions.

2. Literature Review

2.1 Remote sensing for drought studies

Remote sensing has significantly improved drought monitoring by enabling consistent observation of environmental variables over large areas [13]. Advances in optical, thermal, and microwave sensors allow monitoring of precipitation, vegetation condition, soil moisture, and land surface temperature at different spatial and temporal resolutions [2]. These datasets are widely used to derive drought indices and assess drought severity across different climatic regions [14].

Vegetation-based indices such as NDVI, Vegetation Condition Index (VCI), and Vegetation Health Index (VHI) are commonly used to detect vegetation stress [12]. However, these indices are influenced by crop type, soil conditions, and management practices, which limits their reliability when used alone [5]. In addition, vegetation indices often reflect delayed responses to water stress, reducing their usefulness for early drought detection [11].

Soil moisture is a key variable in agricultural drought assessment because it directly represents water availability for crops [7]. Remote sensing techniques, especially microwave-based methods, provide useful estimates of surface soil moisture [26]. However, most datasets represent only near-surface layers and do not capture root-zone soil moisture, which plays a critical role in crop growth and drought development [7]. In addition, the coarse spatial resolution of many soil moisture products limits their use for field-scale analysis [9].

To address these limitations, integrated drought indices combining precipitation, temperature, vegetation, and soil moisture have been developed [10]. Multi-sensor approaches further improve this by combining data from different satellite sources, allowing better representation of drought evolution and crop response [13]. However, integrating multi-source datasets remains challenging due to differences in spatial resolution, temporal frequency, and data uncertainty [9]. Optical sensors are affected by cloud cover, while microwave sensors often have coarse resolution and mixed pixel effects [19]. As a result, current approaches still struggle to provide accurate drought assessment at finer scales [22].

2.2 Need for a multi-sensor remote sensing data integration-based DI

Agricultural drought is multi-dimensional, and a single variable cannot represent its full behavior [14]. Traditional indices based on individual variables fail to capture interactions among climate, soil, and vegetation processes [10]. Even combined indices often rely on fixed weighting schemes, which limits their adaptability across different environmental conditions [13]. This reduces their ability to represent spatial variability and dynamic crop–water relationships.

Multi-sensor remote sensing data integration offers a more comprehensive approach by combining optical, thermal, and microwave observations [2]. This allows simultaneous analysis of vegetation condition, soil moisture, temperature, and precipitation, improving drought characterization [13]. However, most existing approaches still depend on static relationships and are not able to capture non-linear interactions among environmental variables [11].

A key limitation in current drought monitoring is the lack of reliable root-zone soil moisture information and crop growth representation [7]. Surface observations alone are not sufficient, as they do not reflect water availability across soil depths and growth stages [31]. High-resolution and multi-layer soil moisture information is therefore essential, but generating such data remains difficult due to limitations in satellite observations and data integration techniques [7].

Machine learning provides a data-driven solution by enabling modeling of complex and non-linear relationships among multiple variables [3]. These approaches support effective integration of heterogeneous datasets and improve predictive performance [1]. Recent studies show that machine learning models can capture fine-scale spatial variability and temporal drought patterns [9]. However, challenges remain in model robustness and generalization across different agro-climatic regions [18].

Therefore, existing approaches are still not sufficient for reliable field-scale agricultural drought assessment, especially in rain-fed regions. This highlights the need for a data-driven multi-sensor framework that can better capture complex drought dynamics.

3. Materials and Methods

The study examines the Gaya district in the state of South Bihar, India, which has been identified as being drought-prone for several decades. The approach uses multiple satellite-derived datasets for the study area's geographic coordinates (latitude and longitude), and uses a Light Gradient Boosting Machine (LightGBM) model to estimate drought. This model also compares a benchmark to three other models: Random Forest (RF), Extreme Gradient Boosting (XGBoost), and Deep Learning Multi-Layer Perceptron Neural Network (DMLPNN). The entire process includes data collection, preprocessing, indirect downscaling, machine learning NDMI estimation, field validation, and formulation of the Drought Index (DI), which are outlined in the following subsections.

3.1. Study Area

The study area is the Gaya district in the southern part of the state of Bihar, India. The district covers an area of about 4,976 km² and is in a tropical monsoon climate, with an average annual rainfall of about 1,133 mm. Though, the most recent rain fall data from the Tropical Rainfall Measuring Mission (TRMM) records a steady and significant decline in the last twenty years, which is a major cause of persistent cyclic drought in the region [1, 4]. The district does not have canal irrigation, and as a result, farming communities are completely reliant on rain for irrigation [1].

In Gaya, the primary crops grown are rice, wheat, lentils, and some oilseeds. Due to the semi-arid agroclimatic conditions of the region and the continuous decrease of rainfall, agricultural droughts have become more serious and more widespread [1, 5]. All the satellite datasets utilized in this study have been collected and then clipped to the exact longitude–latitude boundaries of Gaya district and then processed to remove all other datasets. A Land Use Land Cover (LULC) map was also created to analyse the study area's spatial distribution of cultivable land, water, rocky, sandy uncultivated, and built-up areas by using the 30 m resolution Landsat-8 multispectral image [10].

3.2 Satellite Datasets and Data Acquisition

For this study, a multi-sensor remote sensing technique was used by integrating data from six different satellite sources. Each data set was chosen based on its documented importance to the study of agricultural droughts as discussed in previous studies [3, 13]. The data extend from May 2017 to May 2023 and cover a 6-year period to study the interannual variability and persistence of droughts [1, 9].

The data for Sentinel-1 Synthetic Aperture Radar (SAR) GRDF was collected with VV and VH polarization channels at a native resolution of 5 m × 20 m and then multilocked to 15 m pixel spacing. Dual polarization SAR backscatter is a good estimation of surface soil moisture based on its sensitivity to the surface's dielectric and its roughness [13, 21].

In addition, the Landsat-8 Operational Land Imager (OLI) and Thermal Infrared Sensor (TIRS) data were used to calculate indices such as the Normalized Differential Moisture Index (NDMI), Normalized Differential Vegetation Index (NDVI), and Land Surface Temperature (LST). The multispectral bands have a resolution of 30 m and the thermal band has a resolution of 100 m [12].

Data on rainfall were obtained from the Tropical Rainfall Measuring Mission (TRMM) with a spatial resolution ~ 27 km resolution (0.25 × 0.25°). Although TRMM data are of low spatial resolution, they provide consistent long-term coverage and have been validated in numerous regional drought studies in South Asia [10]. Data on tropospheric concentrations of methane (CH₄) and ozone (O₃) were obtained from the Sentinel-5P Tropospheric Monitoring Instrument (TROPOMI) with a spatial resolution of 3.5 km and 3.5 km. These two gases are major contributors to radiative forcing and warming of the atmosphere and modelling them enables the

framework to incorporate the large-scale climatic factors of drought [15]. Soil moisture data were obtained from the Soil Moisture Active Passive (SMAP) satellite with a spatial resolution of $36 \text{ km} \times 36 \text{ km}$, and because SMAP provides direct estimates of near-surface soil moisture, it has been validated extensively for agricultural drought studies [7, 9].

3.3. Pre-processing of Spatial Datasets

Given the variations in spatial resolution of the six satellite datasets, a uniform spatial reference systematic pre-processing pipeline was applied so that datasets could be harmonized before evaluation with the machine learning models [11].

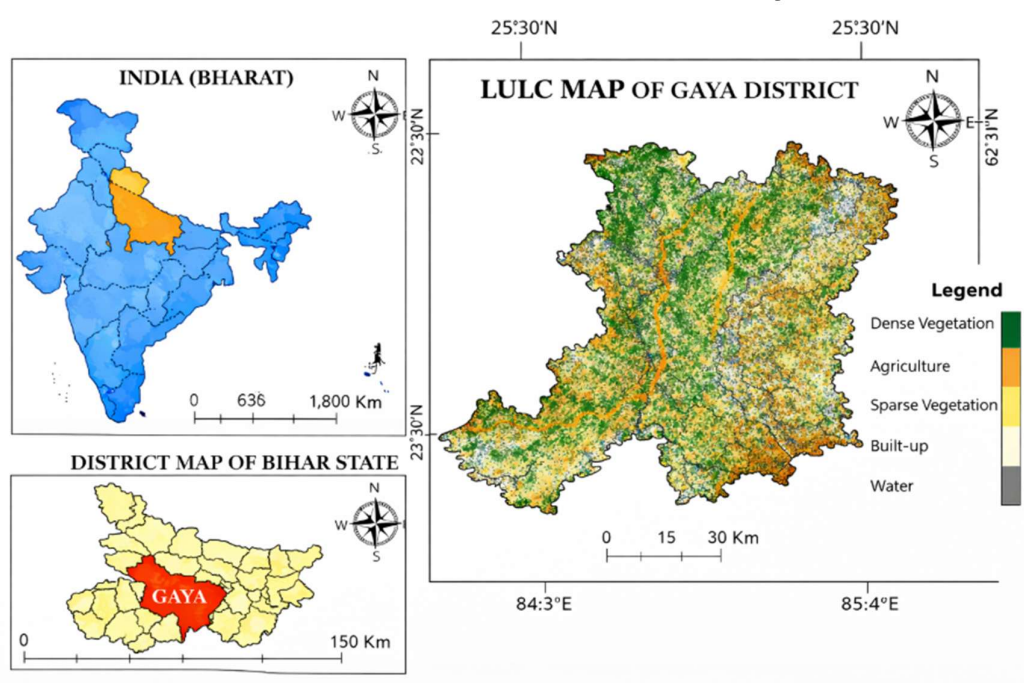


Fig. 1. Study Area Map showing Gaya district in South Bihar.

For the Sentinel-1 SAR datasets, the first step involved radiometric calibration, which transforms raw Digital Number (DN) values into sigma nought (σ^0) backscatter, using the following equation:

$$\sigma^0(dB) = \frac{DN^2}{A^2} \quad \dots (1)$$

where A indicates the β^0 backscatter coefficient and DN the Digital Number value

Following calibration, multilooking was performed with 3 looks in the range direction and 4 looks in the azimuth direction, resulting in $15 \text{ m} \times 15 \text{ m}$ pixel spacing. A Refined Sigma filter was subsequently used to reduce granular speckle noise and then terrain correction was performed, followed by final resampling to a 15 m resolution uniform grid using the bilinear interpolation method [20].

The NDMI was calculated based on the Near-Infrared (NIR) band and the Short-Wave Infrared (SWIR) band on Landsat-8 OLI using the standard formulation [12].

$$NDMI = \frac{(NIR - SWIR)}{(NIR + SWIR)} \quad \dots (2)$$

NIR = Near Infrared band reflectance;

SWIR = Short Wave Infrared band reflectance

NDVI was calculated similarly, using NIR and the Red band of the Landsat-8 imagery [12]:

$$NDVI = \frac{(NIR - R)}{(NIR + R)} \quad \dots (3)$$

R = reflectance of Red band

Both index maps were resampled to 15 m resolution to align spatially with the Sentinel-1 SAR data. To avoid substantial information loss, we used the methodology described in [11] to determine the direct sampling of the TRMM, SMAP, and Sentinel-5P data that are available at coarse spatial resolutions covering the entire study area in one pixel. We extracted fortnightly average values (2 per month) from each of the coarse resolutions and input them into an Ordinary Least Squares (OLS) regression model with the finer resolution SAR, LST, NDMI, and NDVI. Per-pixel estimates of precipitation, soil moisture, and the concentration of greenhouse gases were then produced at 15 m spatial resolution for the entire study period (2017–2023) using this OLS-based indirect downscaling technique [13].

3.4. Machine Learning Approaches

In machine learning models, the aim is to estimate NDMI, which is used to understand soil moisture conditions in the Gaya district. The models are trained to understand the NDMI values in relation to the nine input variables that were derived from the nine input parameters. The data is in the form of a nonlinear regression task. The data is inputted in per pixel form, for data intervals of 15 m, covering the time span of 2017-05 to 2023-04.

The models to be trained are LightGBM, Random Forest, XGBoost, and DMLPNN, which are used in the study as comparative models [3, 18].

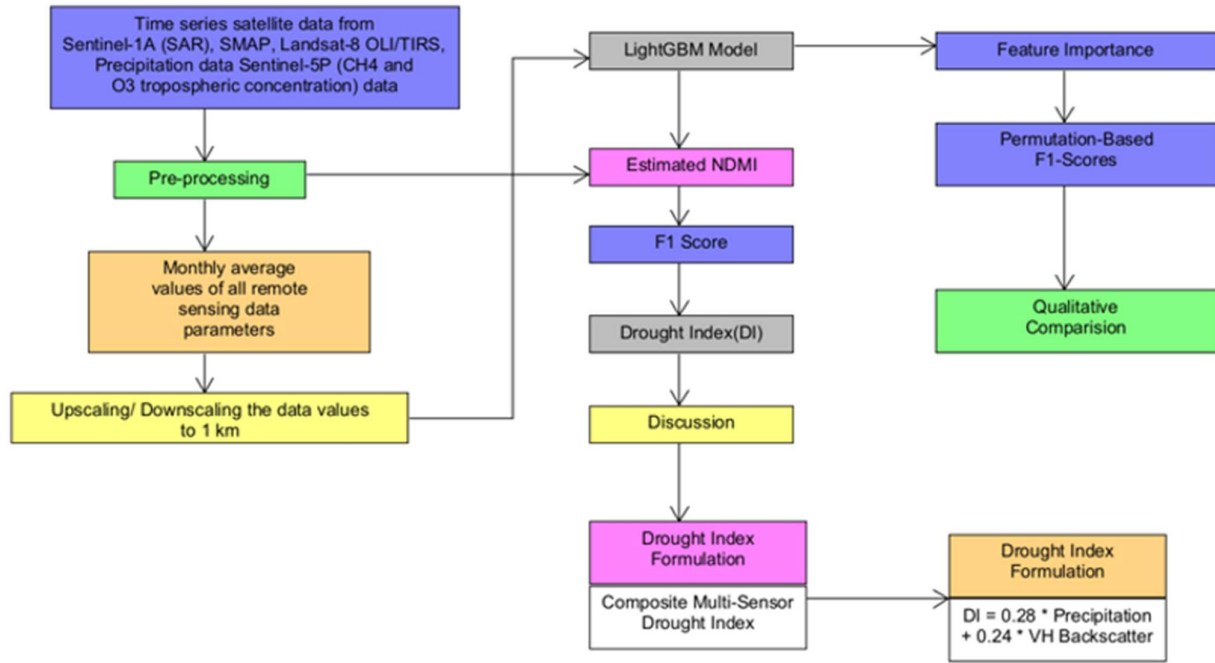


Fig.2. Overall Methodology Flow Diagram

3.4.1. Primary Model

3.4.1.1. LightGBM Model

LightGBM is the name of the model chosen to be the most predictive of the ones used in the study. Such a choice is based on the fact that it has the greatest computational efficiency, especially when it comes to remote sensing regression problems, and when it comes to large and complex datasets [3]. LightGBM is a type of gradient boosting machine, and what that means is that it builds an ensemble of decision tree models over time, and each tree corrects the faults that the previous tree has made. Unlike other models that are considered gradient boosting machines, LightGBM has a strategy of growing leaf nodes and it is realized through a combination of GOSS and Exclusive Feature Bundling (EFB), which means that the whole process is made faster, and consumes less computational resources, while producing more accurate results [18]. The model was set with the following key hyperparameters, tuned through the use of Cross validation: number of estimators = 500, learning rate = 0.05, maximum depth = -12 (leaf-wise), number of leaves = 63, minimum child samples = 20, and subsample = 0.8. The nine satellite-derived parameters (SAR VV backscatter SAR VH backscatter, LST, NDVI, NDMI, TRMM precipitation, SMAP soil moisture, CH₄ concentration and O₃ concentration) comprised the input feature set and the estimated NDMI was used as the target variable. A 70:30 train-test split was uniformly applied across all the models as to maintain consistency [16].

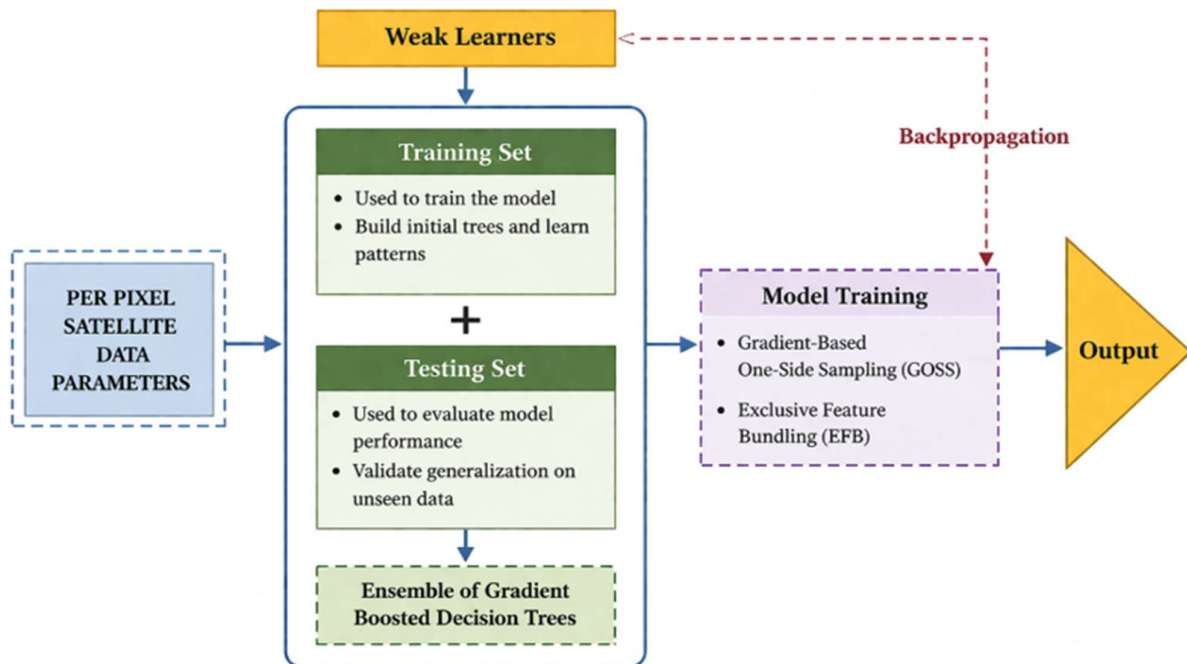


Fig.3. LightGBM Model Architecture

3.4.2. Comparative Model

3.4.2.1. Random Forest Model

Random Forest (RF) is an ensemble learning method that constructs a large number of decision trees during training and outputs the mean prediction across all trees for regression tasks. RF introduces randomness both through bootstrap sampling of the training data and through random feature selection at each split node, which together reduce variance and improve generalization [11]. In this study, the RF model was configured with 500 estimators and a maximum feature subset of the square root of the total number of features. RF has been widely applied in remote sensing-based drought and soil moisture estimation and serves here as a well-established baseline for comparison with LightGBM.

3.4.2.2. XGBoost Model

Extreme Gradient Boosting (XGBoost) is another tree-based boosting framework that sequentially fits models to minimize an objective function combining a loss term and a complexity penalty. XGBoost employs a level-wise tree growth strategy with second-order gradient approximations, which improves convergence stability compared to standard gradient boosting [15]. The model was configured with 500 estimators, a learning rate of 0.05, maximum tree depth of 6, and a subsample ratio of 0.8. XGBoost is frequently used as a competitive benchmark in machine learning-based geospatial modelling, making it an appropriate comparison model for this study.

3.4.2.3. DMLPNN Model

The Deep Learning Multi-Layer Perceptron Neural Network (DMLPNN) is included as a third comparative model to represent deep learning-based approaches to drought estimation. The network architecture consists of an input layer with nine nodes, seven hidden layers (four layers of 40 nodes and three layers of 50 nodes each), five activation layers using the Rectified Linear Unit (ReLU) function, and five dropout layers to mitigate overfitting. The output layer produces a single continuous NDMI value. The DMLPNN has previously demonstrated strong performance in multi-sensor remote sensing regression tasks and serves as a deep learning benchmark in this comparative analysis.

Table 1. Dataset Description

S. No.	Variable	Description	Unit/Range	Time Period
1.	year	Year of Observation	2018 - 2022	Jan – 2018 to Dec - 2022
2.	month	Month of Observation	1 - 12	Jan – 2018 to Dec - 2022
3.	ndvi	Normalized Difference Vegetation Index – derived from Landsat-8 OLI (Red & NIR bands)	-1 to 1 (obs : ~0.19 – 0.48)	Jan – 2018 to Dec - 2022
4.	lst	Land Surface Temperature – derived from Landsat-8 thermal band	°C (obs: ~20 – 43)	Jan – 2018 to Dec - 2022
5.	precip	Rainfall – from TRMM dataset (temporal res :every 3hrs, 15-day avg)	mm (obs: ~23 – 106)	Jan – 2018 to Dec - 2022
6.	smap	Soil Moisture – from SMAP Active Passive Satellite (temporal res: every 49 min, 15-day avg)	cm ³ /cm ³ (obs: ~2.2 – 6.8)	Jan – 2018 to Dec - 2022
7.	vv	Sentinel-1 SAR VV polarization backscatter (GRDH product, multilook to 15m)	dB (obs: ~ -16 to -7)	Jan – 2018 to Dec - 2022
8.	vh	Sentinel-1 SAR VH polarization backscatter (GRDH product, multilook to 15m)	dB (obs: ~ -23 to -13)	Jan – 2018 to Dec - 2022
9.	vv_vh_ratio	Ratio of VV to VH SAR backscatter – dual-polarization indicator	Unitless (obs: ~0.53 – 0.68)	Jan – 2018 to Dec - 2022
10.	Ch4	Tropospheric CH4 concentration – from Sentinel-5P (every 8 hrs, 15-day avg)	ppb (obs: ~ 1880 – 1920)	Jan – 2018 to Dec - 2022
11.	ndmi	Normalized Difference Moisture Index – derived from Landsat-8 NIR & SWIR bands	-1 to 1 (obs: ~ -0.18 to 0.01)	Jan – 2018 to Dec - 2022

3.5. Performance Metrics

The performance of all four models was evaluated using a consistent 70:30 train-test split. Three standard statistical metrics were used for accuracy assessment: the Coefficient of Determination (R^2), Root Mean Square Error (RMSE), and Mean Absolute Error (MAE). The R^2 statistic, which measures the proportion of variance in the observed data explained by the model, was computed as:

3.5.1. R^2 Coefficient:

The Coefficient of Determination (R^2) measures how well a model explains the variability of the observed data. It represents the proportion of variance in the dependent variable that is predictable from the independent variables.

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \quad \dots (4)$$

Where, y_i denotes observed values, $\hat{y}_i$ be the predicted values and $\bar{y}$ be the mean of observed values.

3.5.2. Mean Square Error:

Mean Squared Error (MSE) represents the average squared difference between observed and predicted values, providing a measure of prediction accuracy with higher sensitivity to large errors.

$$MSE = \left(\frac{1}{n}\right) * \sum (y_i - \hat{y}_i)^2 \quad \dots (5)$$

3.5.3. Root Mean Square Error:

The RMSE quantifies the average magnitude of prediction errors and was calculated as:

$$RMSE = \sqrt{\left[\frac{\sum (y_i - \hat{y}_i)^2}{n} \right]} \quad \dots (6)$$

n = total number of observations

These three metrics together provide a comprehensive assessment of each model's predictive accuracy, error magnitude, and robustness. The LightGBM model's performance was compared directly against RF, XGBoost, and DMLPNN across all three metrics to identify the best-performing model for NDMI estimation in the Gaya district.

3.6. Land Use/Land Cover (LULC) Masking

To restrict the drought analysis to agricultural lands, all non-agricultural LULC features including urban built-up zones, water bodies, rocky terrain, and fallow lands were masked from the output maps. An NDVI-based masking approach was applied, wherein the difference between NDVI maps from the pre-harvest and post-harvest periods of both the Kharif (rice) and Rabi (wheat) crop seasons was computed. Since the spatial extent of agricultural fields remains consistent between seasons, one of the NDVI difference layers was converted into a vector polygon mask. In ArcGIS this was applied to clip the estimated NDMI and DI output maps, retaining only agricultural pixels for the drought assessment.

3.7. Feature Importance (FI) Score Computation

To quantify the individual contribution of each input parameter to the NDMI estimation by the LightGBM model, a permutation-based Feature Importance (FI) score was computed. This method is model-agnostic and does not rely on any assumptions about the internal structure of the model, making it particularly suitable for nonlinear and ensemble-based approaches [3]. The technique involves randomly shuffling the values of one input feature at a time during prediction, while keeping all other features unchanged. The resulting degradation in prediction accuracy (measured by RMSE) quantifies the importance of the shuffled feature. A larger increase in RMSE upon shuffling indicates a greater contribution of that parameter to the model's predictive performance [9, 18]. The FI scores for all nine parameters were normalised and expressed as percentage contributions. These weights were then directly used in the formulation of the novel multi-sensor Drought Index (DI), ensuring that the most influential parameters carry the greatest weight in the final drought map [11].

3.8. Drought Index (DI) Formulation

A novel multi-sensor Drought Index (DI) was developed for the Gaya district using the FI scores derived from the LightGBM model as the weighting coefficients for each input parameter. This approach of constructing a composite drought index from weighted satellite-derived parameters is well established in the literature and addresses the major limitation of traditional single-parameter indices such as the Palmer Drought Severity Index (PDSI) and the Standardized Precipitation Index (SPI), which fail to capture the multidimensional nature of agricultural drought.

Based on the percentage contribution of each parameter as determined by the FI score, the DI for the Gaya district was computed as follows:

$$DI = \frac{(0.0795 * LST) + (0.24 * ndvi) + (0.0597 * SMAP) + (0.0688 * VV) + (0.0856 * VH) + (0.011 * VV_{VH_{Ratio}}) + (0.01068 * CH_4)}{100} \dots (7)$$

where each coefficient represents the normalized FI score (as a fraction of 100) of the respective parameter.

The DI map was generated for the entire Gaya district at 15 m spatial resolution for the year 2022 and subsequently masked to retain only agricultural land pixels as described in Section 2.7. To track the temporal evolution of drought severity, DI values were also computed for each year from 2018 to 2022 using the corresponding archived satellite datasets. Annual maximum and minimum DI values were tabulated to assess the progressive trend in drought intensity over time [6, 9].

For qualitative validation, the 2019 DI map was visually compared with the NDVI difference map (May 2018 vs. May 2022) and with the LightGBM-estimated NDMI map for the same period. Spatial agreement between regions of high DI values and areas showing declining vegetation and reduced soil moisture would confirm the physical consistency and reliability of the proposed Drought Index [13].

3.9. Role of CH₄ and O₃ in Influencing Land Surface Temperature

The inclusion of tropospheric CH₄ and O₃ concentrations as input parameters is grounded in their well-established roles as greenhouse gases responsible for radiative forcing and atmospheric warming. CH₄ absorbs long-wave terrestrial radiation, amplifying surface-level heating, while tropospheric O₃ intercepts both incoming solar radiation and outgoing terrestrial radiation, producing net warming near the land surface [15]. Since increasing Land Surface Temperature accelerates evapotranspiration and reduces the soil's capacity to retain moisture, incorporating greenhouse gas concentrations as model inputs allows the LightGBM framework to account for the large-scale climatic drivers of agricultural drought — something that purely precipitation- or vegetation-based indices inherently overlook [3, 5].

This multi-dimensional approach aligns closely with the growing consensus in the remote sensing and climatology literature that robust agricultural drought assessment must simultaneously consider land-surface processes, atmospheric forcing, vegetation response, and soil moisture dynamics [11, 13]. By integrating all of these dimensions into a single LightGBM-based modelling framework, this study advances beyond the limitations of conventional single-sensor drought indices and offers a more holistic and physically grounded tool for district-level agricultural drought monitoring [19, 20].

4. Experimental results

The average annual rainfall of Gaya district for the period 2000 to 2023 was extracted from TRMM data and is presented in Fig. 4. The results show noticeable inter-annual fluctuations throughout the study period. Several years recorded rainfall below the district's long-term average of approximately 1,133 mm. Recurring below-normal rainfall years were observed across the two-decade period, interspersed with relatively higher rainfall years.

Annual mean LST maps generated from Landsat-8 TRS data for the period 2017 to 2022 are presented in Fig. 5. The maps show a general increasing trend in land surface temperature across Gaya district over the study period. A marginal decrease in mean LST was recorded in 2020 compared to the preceding year. The spatial distribution of LST shows higher temperature values concentrated in the southern and central portions of the district, with relatively lower temperatures observed in the north.

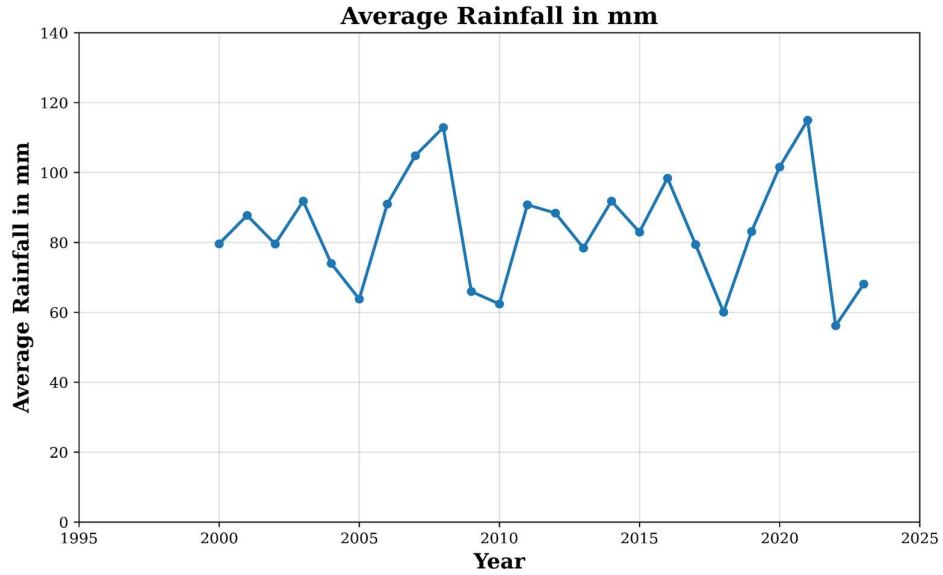


Fig. 4. Average Annual Rainfall of Gaya district South Bihar

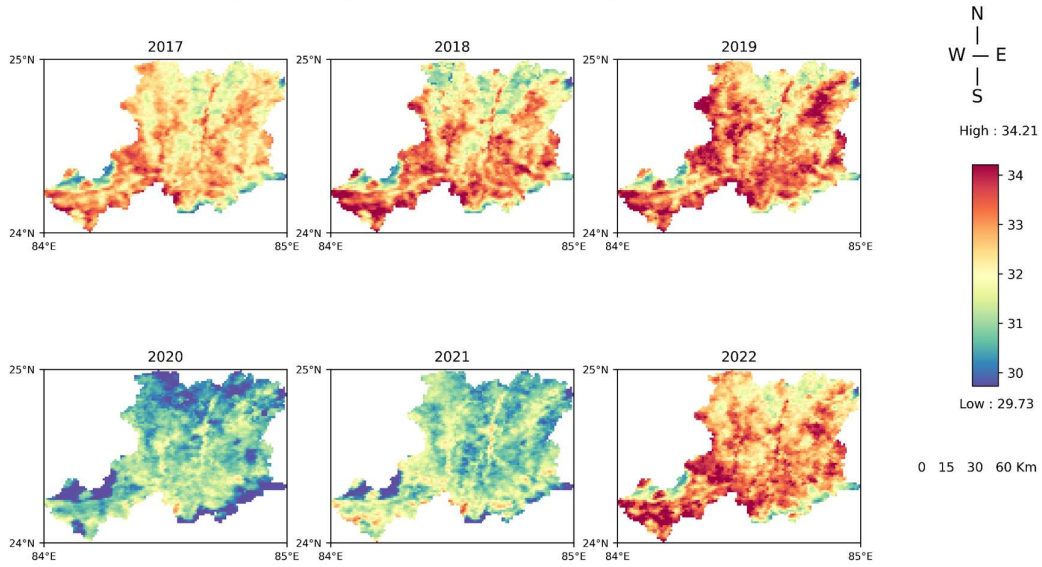


Fig. 5. Mean Annual LST maps of Gaya district from 2017 to 2022(in Celsius)

The spatial distribution of NDMI estimated by the LightGBM model for Gaya district is shown in Fig. 6. NDMI values across the district ranged from negative values in the southern portions to moderate positive values in the northern and eastern zones. The southern part of the district consistently exhibited lower NDMI values compared to the other areas. The estimated NDMI map was generated at 15m spatial resolution and masked to retain only agriculture pixels.

The NDVI difference map computed between May 2017 and May 2022 is presented in Fig. 7. Negative NDVI difference values are concentrated predominantly into the southern and south-eastern parts of Gaya district. The northern and north-eastern portions of the district show comparatively smaller negative or near-zero difference values. Areas with lower NDMI values in Fig. 6. Correspond spatially with areas of greater NDVI decline in Fig. 7.

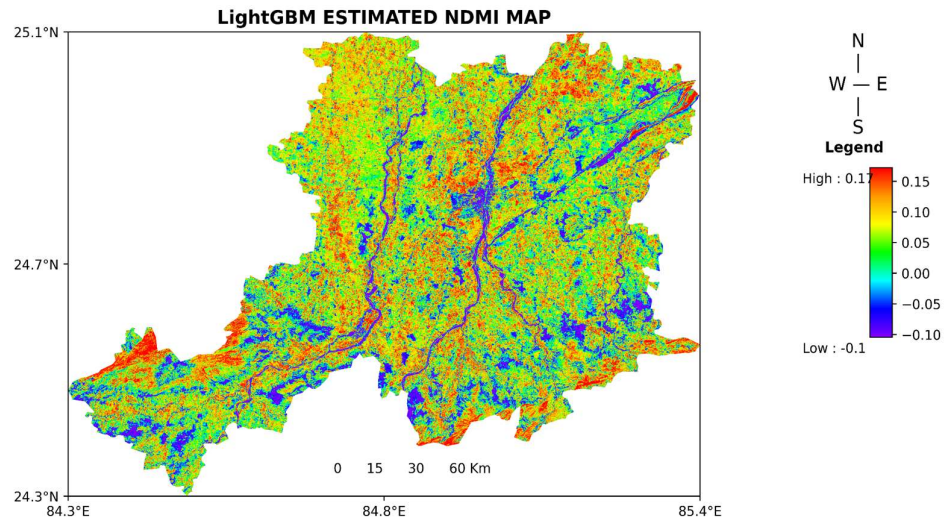


Fig .6. LightGBM estimated NDMI map for Gaya district

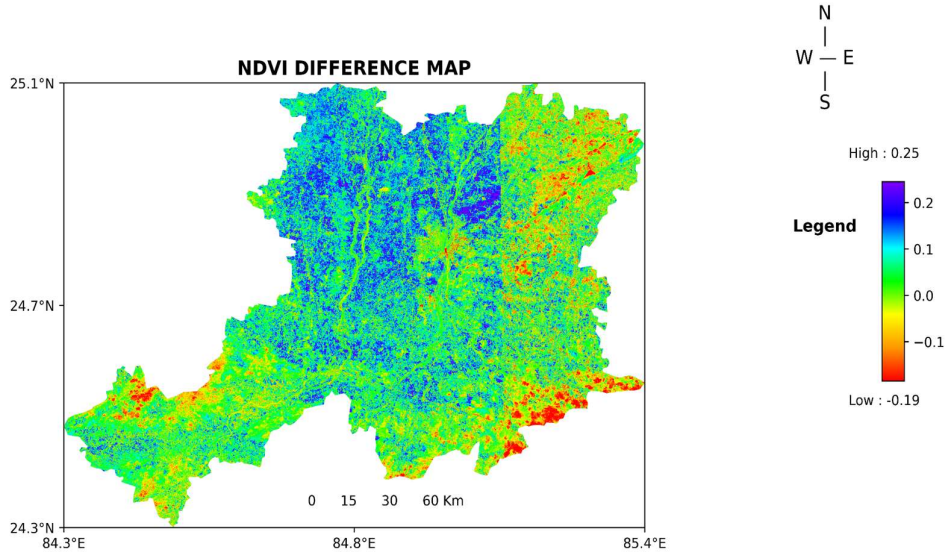


Fig .7. NDVI Difference map of year 2017 and 2023

The permutation-based Feature Importance (FI) scores derived from the LightGBM model are presented in Fig. 8. NDVI recorded The highest FI score followed by VV/VH ratio, CH4, VH backscatter, LST, VV backscatter and SMAP. These FI scores were used as weighting coefficients in the Drought index formulation as given in Equation 7.

The DI map generated for Gaya district for the year 2019 is presented in Fig. 9. Higher DI values are concentrated in the southern And south-eastern portions of the district. The northern regions show comparatively lower DI values. The annual DI values computed for the period 2018 to 2022 are tabulated in table 2. Maximum DI values also show a progressive increase from 2018 to 2022, with 2018 recording the highest maximum DI value across the study period. Minimum DI values also show an upward shift over the same period. The spatial distribution of validation sample points overlaid on the DI map is shown in Fig. 10. The validation points are distributed across low, moderate, and high drought severity zones across the district. Points located in the southern and south-eastern regions correspond to areas of higher DI values in the drought index map. Spatial agreement is observed between the DI map (Fig. 9), the NDMI map (Fig. 6), and the NDVI difference map (Fig. 7), with regions of high drought severity consistently identified across all three outputs.

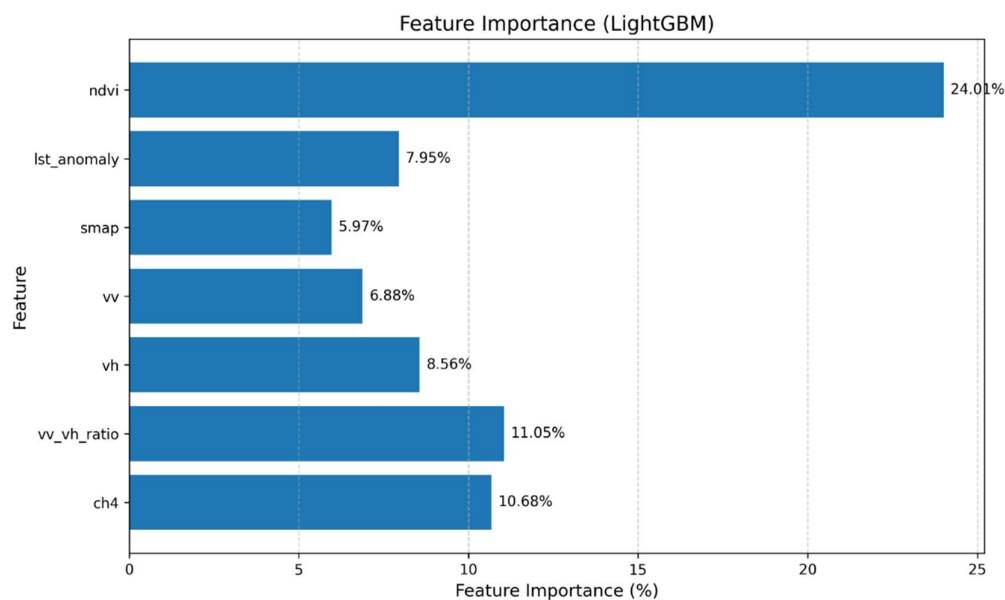


Fig. 8. LightGBM feature importance (%) for NDMI estimation.

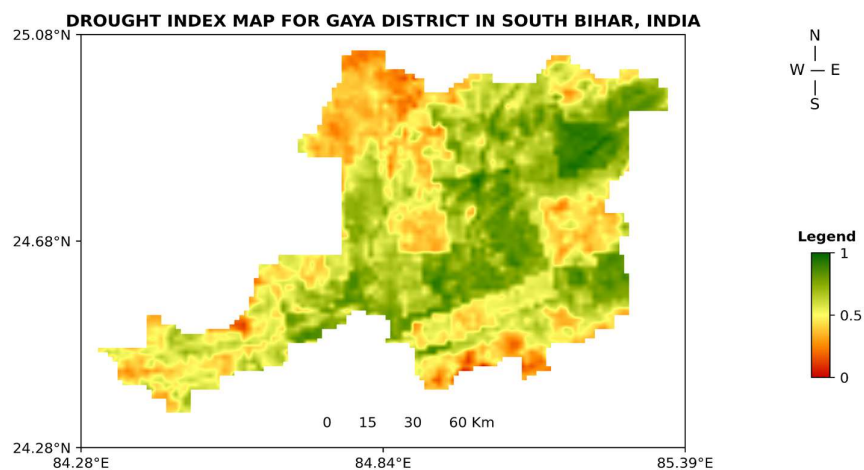


Fig. 9. DI map of the year 2019 for Gaya district

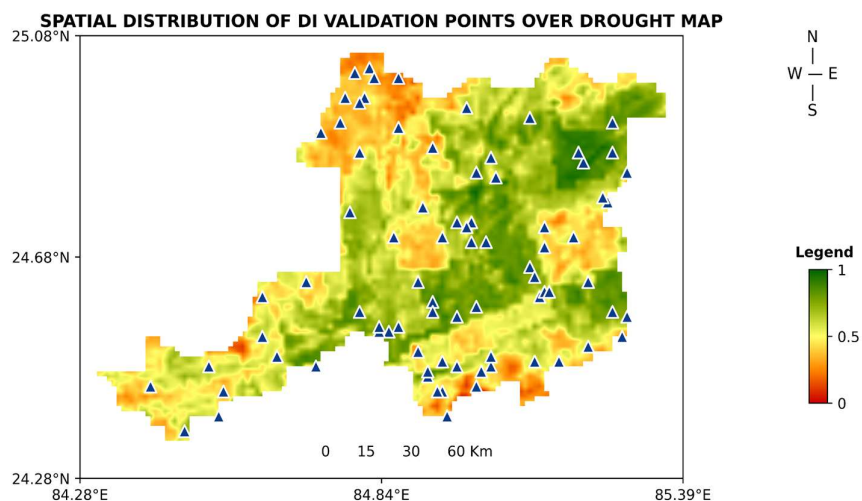


Fig.10. Spatial distribution map of drought index calculation

Table 2. Time series Annual DI values from 2018 - 2022

S. No	DI (year)	MAXIMUM VALUE	MINIMUM VALUE
1	2018	0.98	0.11
2	2019	0.89	0.24
3	2020	0.69	0.18
4	2021	0.63	0.15
5	2022	0.65	0.14

4.1. Discussion

The LightGBM model achieved an R^2 of 0.89 on the test data, which is higher than results reported in similar studies. Zhang et al. [2] reported R^2 values between 0.75 and 0.82 for a drought index study over semi-arid croplands using vegetation, Temperature, and soil moisture data. Sanchez et al. [3] obtained an R^2 of around 0.78 using MODIS satellite data for drought monitoring. In the present study, the inclusion of Sentinel-1 SAR backscatter data in both VV and VH polarizations contributed to the improved accuracy, as SAR data can reliably detect soil moisture and vegetation changes even under cloudy conditions. Furthermore, CH₄ concentration data from Sentinel-5P was incorporated as an additional input parameter, which provides atmospheric forcing information that earlier studies has not considered.

CH₄ absorbs outgoing long-wave terrestrial radiation, while tropospheric O₃ intercepts both incoming solar radiation and outgoing terrestrial radiation, together producing a net warming effect at the land surface that drives additional moisture loss from the soil. By including these greenhouse gas inputs, the LightGBM model captures a slowly varying but physically meaningful thermal forcing signal that improves NDMI estimation in months when precipitation and LST inputs are spatially ambiguous across the district.

Traditional single-variable drought indices such as the Palmer Drought Severity Index (PDSI) and the Standardized Precipitation Index (SPI) rely on fixed meteorological inputs and do not directly represent soil moisture availability or vegetation water stress, which limits their applicability at fine spatial scales in heterogeneous rain-fed agricultural landscapes. These indices also depend on sparse station networks with uneven spatial coverage, making them unsuitable landscapes. These indices also depend on sparse station networks with uneven spatial coverage, making them unsuitable for block-level drought differentiation within a district the size of Gaya. Vegetation-based indices such as VCI and VHI respond to canopy stress rather than soil moisture directly and are known to exhibit delayed responses to water deficit, reducing their utility for early drought detection. In addition, their sensitivity to crop type, soil background, and management practices introduces variability that limits reliability when a single index is used without additional parameters.

The data-driven weighting scheme employed in this study where LightGBM permutation FI scores directly determine the DI coefficients contrasts with the fixed-weight approaches of earlier composite indices and addresses the primary limitation identified across studies. By deriving weights from the actual predictive contribution of each variable to NDMI variability in Gaya district, the DI coefficients reflect local environmental relationships rather than climatological assumptions imported from other regions or seasons. The resulting DI spatially identifies the southern and south-eastern blocks of Gaya district as the most severely drought-affected zones across the 2018–2022 period, with the 2019 DI map showing the most widespread high-severity pattern, consistent with known below-normal monsoon rainfall in that year. These spatial patterns are independently matched by the LightGBM-estimated NDMI map (Fig. 6), the NDVI difference map (Fig. 7), and the distribution of validation sample points across drought severity classes in Fig. 10, confirming the physical consistency of the proposed index.

5. Conclusion

This study developed a machine learning-based multi-sensor framework for agricultural drought assessment in the Gaya district, where agriculture is strongly dependent on rainfall due to the absence of canal irrigation. The aim was to reduce dependence on field data while improving the reliability of drought assessment using satellite observations. By combining multiple remote sensing datasets, the study provides a practical approach for monitoring drought conditions in data-scarce regions.

A major challenge was the difference in spatial and temporal resolution among the datasets. This was handled through a consistent data processing approach, allowing SAR backscatter, vegetation and moisture indices, land surface temperature, precipitation, soil moisture, and atmospheric parameters to be used together within a single framework. The results show that freely available satellite datasets can be effectively integrated for district-level drought assessment.

Among the evaluated models, LightGBM showed the best performance for NDMI estimation, with high prediction accuracy and stable results. This confirms that machine learning can capture complex relationships among drought-related variables. Feature Importance analysis further helped identify the contribution of each variable and supported the development of a multi-sensor Drought Index (DI).

The DI clearly captured spatial differences in drought conditions, with higher drought severity observed in the southern and south-eastern parts of the study area. These patterns matched variations in vegetation condition and moisture levels, showing that the DI represents drought conditions in a realistic way.

The main outcomes of the study are:

- Integration of multi-sensor datasets with different resolutions for district-level drought assessment.
- Reliable estimation of NDMI using machine learning models.
- Development of a multi-sensor Drought Index (DI) without reliance on field data.
- A data-driven framework for drought assessment using only freely available satellite datasets.
- Reduced dependence on ground data, making the approach time-efficient and cost-effective.
- A scalable method that can be applied to other rain-fed and drought-prone regions.

Overall, this study shows that combining multi-sensor remote sensing data with machine learning improves drought assessment compared to traditional methods. The proposed framework provides a reliable and practical solution for monitoring drought conditions in regions where ground data are limited.

5.2 Future Work

The proposed framework can be further improved by incorporating additional data sources and validation strategies to enhance its accuracy and applicability.

- Validation using ground-based soil moisture data to assess the reliability of NDMI and the developed Drought Index (DI).
- Integration of evapotranspiration datasets to better represent crop water stress conditions.
- Application of advanced deep learning and time-series models to improve prediction of drought evolution.
- Extension of the framework to larger spatial scales for regional monitoring and drought forecasting.

References

- [1] Kumar, P., Chandrakant, S.C., Ranjan, S., Tripathi, A., 2025. Agricultural drought assessment using deep learning and multi-sensor remote sensing data integration. *Phys. Chem. Earth* 140, 104006. <https://doi.org/10.1016/j.pce.2025.104006>
- [2] Kumar, P., Tripathi, A., Moniruzzaman, M., Chandrakant, S.C., 2025. Multi-sensor remote sensing data integration for agricultural droughts: a PRISMA-based review in the Indian context. *Earth Syst. Environ.* 9, 2699–2722. <https://doi.org/10.1007/s41748-024-00540-8>
- [3] Mondal, S., Prasad, K.A., Achu, A.L., Kaliraj, S., Balasubramani, K., 2026. Advancements in spatio-temporal agricultural drought monitoring and modeling: a comprehensive review on multi-source remote sensing and machine learning techniques. *Theor. Appl. Climatol.* 157, 222. <https://doi.org/10.1007/s00704-026-06136-8>
- [4] Panda, K.C., Paramaguru, P.K., Singh, R.M., Singh, S.K., 2026. Assessing meteorological drought indices for monitoring agricultural drought using SPEI: a remote sensing approach. *J. Atmos. Sol.-Terr. Phys.* 279, 106726. <https://doi.org/10.1016/j.jastp.2026.106726>
- [5] Rahman, G., Khalid, S., Arshad, S., Moazzam, M.F.U., Kwon, H.-H., 2025. Remote sensing-based spatiotemporal assessment of agricultural drought and its impact on crop yields in Punjab, Pakistan. *Sci. Rep.* 15, 06095. <https://doi.org/10.1038/s41598-025-06095-6>
- [6] Bozorgi, M., Cristóbal, J., Casadesús, J., 2026. Assessing the performance of multi-timescale drought indices for monitoring agricultural drought impacts on wheat yield. *Agric. Water Manag.* 323, 110092. <https://doi.org/10.1016/j.agwat.2025.110092>
- [7] Eshetie, H.A., Alamirew, T., Abera, A., Sahu, D., Ayele, A.D., 2026. High-resolution root-zone soil moisture for agricultural drought assessment using Sentinel-2 and hybrid modeling in the Lake Tana Basin, Ethiopia. *Remote Sens. Appl.: Soc. Environ.* 41, 101900. <https://doi.org/10.1016/j.rsase.2026.101900>
- [8] Duran, S., Doğan, M., 2025. Agricultural drought survey based upon multiple indices in Southeast Anatolia Project Region, Upper Mesopotamia, Türkiye. *J. Indian Soc. Remote Sens.* 54, 123–140. <https://doi.org/10.1007/s12524-025-02261-5>
- [9] Wu, Y., Zhu, L., Ding, M., Shi, L., 2026. Multi-dimensional monitoring of agricultural drought at the field scale. *Agriculture* 16(2), 227. <https://doi.org/10.3390/agriculture16020227>
- [10] Kulkarni, S.S., Wardlow, B.D., Bayissa, Y.A., Tadesse, T., Svoboda, M.D., Gedam, S.S., 2020. Developing a remote sensing-based combined drought indicator approach for agricultural drought monitoring over Marathwada, India. *Remote Sens.* 12(13), 2091. <https://doi.org/10.3390/rs12132091>
- [11] Mullapudi, A., Vibhute, A.D., Mali, S., Patil, C.H., 2023. A review of agricultural drought assessment with remote sensing data: methods, issues, challenges and opportunities. *Appl. Geomat.* 15, 1–13. <https://doi.org/10.1007/s12518-022-00484-6>
- [12] Huete, A.R., Didan, K., van Leeuwen, W., Miura, T., Glenn, E., 2010. MODIS vegetation indices. In: *Land Remote Sensing and Global Environmental Change*. Springer, pp. 579–602. https://doi.org/10.1007/978-1-4419-6749-7_26
- [13] Zhang, X., Chen, N., Li, J., Chen, Z., Niyogi, D., 2017. Multi-sensor integrated framework and index for agricultural drought monitoring. *Remote Sens. Environ.* 188, 141–163. <https://doi.org/10.1016/j.rse.2016.10.045>
- [14] Sánchez, N., González-Zamora, Á., Martínez-Fernández, J., Piles, M., Pablos, M., 2018. Integrated remote sensing approach to global agricultural drought monitoring. *Agric. For. Meteorol.* 259, 141–153. <https://doi.org/10.1016/j.agrformet.2018.04.022>
- [15] Zhang, T., 2023. Drought monitoring and forecasting using remote sensing and machine learning. In: *Advances in Remote Sensing*. Elsevier, pp. 101–125. <https://doi.org/10.1016/B978-0-443-38345-8.00010-7>
- [16] Huang, Y., Liu, Y., Shi, R., Ren, H., 2023. Application of remote sensing and GIS in drought and flood assessment and monitoring. *Water* 15(3), 541. <https://doi.org/10.3390/w15030541>
- [17] Ahmadi, M., Ramezani Etedali, H., Kaviani, A., Tavakoli, A., 2024. Evaluation of meteorological drought indices using remote sensing. *J. Atmos. Sol.-Terr. Phys.* 265, 106387. <https://doi.org/10.1016/j.jastp.2024.106387>
- [18] Bojer, A.K., Biru, B.H., Al-Quraishi, A.M.F., Debele, T.G., Negera, W.G., Woldeasilasie, F.F., Esubalew, S.Z., 2024. Machine learning and remote sensing-based time series analysis for drought risk prediction in Borena Zone, Southwest Ethiopia. *J. Arid Environ.* 229, 105160. <https://doi.org/10.1016/j.jaridenv.2024.105160>
- [19] Leisenheimer, L., Wellmann, T., Jänicke, C., Haase, D., 2024. Monitoring drought impacts on street trees using remote sensing: disentangling temporal and species-specific response patterns with Sentinel-2 imagery. *Ecol. Inform.* 83, 102659. <https://doi.org/10.1016/j.ecoinf.2024.102659>
- [20] Dos Anjos de Souza, R., Moura, V., Paloschi, R.A., Aguiar, R.G., Webler, A., Borma, L.S., 2022. Assessing drought response in the Southwestern Amazon Forest by remote sensing and in situ measurements. *Remote Sens.* 14(7), 1733. <https://doi.org/10.3390/rs14071733>
- [21] Elsaid, A., Yimer, E.A., Mogheir, Y., Huysmans, M., Villani, L., van Griensven, A., 2025. Groundwater drought and anthropogenic amplifiers: a review of assessment and response strategies in arid and semi-arid areas. *Sci. Total Environ.* 978, 179406. <https://doi.org/10.1016/j.scitotenv.2025.179406>
- [22] Kotzur, I., Moore, B.D., Meakin, C., Evans, M.J., Youngentob, K.N., 2024. Identifying habitat productivity thresholds to assess the effects of drought on a specialist folivore. *Remote Sens.* 16(17), 3279. <https://doi.org/10.3390/rs16173279>
- [23] Lai, Y., Tang, H., Zhan, C., Hong, S., Ran, Q., 2025. Evaluating the cumulative and time-lag effects of vegetation response to drought in the Lancang–Mekong River Basin. *Ecol. Indic.* 114113. <https://doi.org/10.1016/j.ecolind.2025.114113>
- [24] Rockstad, G.B.G., Austin, R.E., Gouveia, B.T., Carbajal, E.M., Milla-Lewis, S.R., 2023. Assessing unmanned aerial vehicle-based imagery for breeding applications in St. Augustinegrass under drought and non-drought conditions. *Crop Sci.* <https://doi.org/10.1002/csc2.21128>

- [25] Duarte, E., Rubilar, R., Matus, F., Garrido-Ruiz, C., Merino, C., Smith-Ramirez, C., Aburto, F., Rojas, C., Stehr, A., Dörner, J., Nájera, F., Barrientos, G., Jofré, I., 2024. Drought and wildfire trends in native forests of South-Central Chile in the 21st century. *Fire* 7(7), 230. <https://doi.org/10.3390/fire7070230>
- [26] En-Nagré, K., Aqnouy, M., Ouarka, A., Naqvi, S.A.A., Bouizrou, I., El Messari, J.E.S., Tariq, A., Soufan, W., Li, W., El-Askary, H., 2024. Assessment and prediction of meteorological drought using machine learning algorithms and climate data. *Clim. Risk Manag.* 100630. <https://doi.org/10.1016/j.crm.2024.100630>
- [27] Anderson, M.C., Hain, C., Wardlow, B., Pimstein, A., Mecikalski, J.R., Kustas, W.P., 2011. Evaluation of drought indices based on thermal remote sensing of evapotranspiration over the continental United States. *J. Clim.* 24, 2025–2044. <https://doi.org/10.1175/2010JCLI3812.1>
- [28] Aziz, M.A., Moniruzzaman, M., Tripathi, A., Hossain, M.I., Ahmed, S., Rahaman, K.R., Rahman, F., Ahmed, R., 2022. Delineating flood zones using synthetic aperture radar data for the 2020 flood in Bangladesh. *Earth Syst. Environ.* 6, 733–743. <https://doi.org/10.1007/s41748-022-00295-0>
- [29] Bhuiyan, C., Singh, R.P., Kogan, F.N., 2006. Monitoring drought dynamics in the Aravalli region (India) using indices based on ground and remote sensing data. *J. Appl. Geophys.* 57, 200–215. <https://doi.org/10.1016/j.jag.2006.03.002>
- [30] O’Brien, L.V., Berry, H.L., Coleman, C., Hanigan, I.C., 2014. Drought as a mental health exposure. *Environ. Res.* 131, 181–187. <https://doi.org/10.1016/j.envres.2014.03.014>
- [31] Seneviratne, S.I., Corti, T., Davin, E.L., Hirschi, M., Jaeger, E.B., Lehner, I., Orlowsky, B., Teuling, A.J., 2010. Investigating soil moisture–climate interactions in a changing climate: a review. *Earth-Sci. Rev.* 99, 125–161. <https://doi.org/10.1016/j.earscirev.2010.02.004>
- [32] Irawan, L.Y., Putri Wahyu, H.Z., Firdaus, A.L., Wijaya, A.A., Azizah, V., Prasetyo, W.E., 2024. Erosion prediction based on terrestrial survey and RUSLE method in Precet Forest Park, Malang. *Geosfera Indones.* 9(3), 394–417. <https://doi.org/10.19184/geosi.v9i3.45479>
- [33] Li, N.; Li, S.; Zhao, B.; Yuan, X.; Zhu, J. *Global Agricultural Drought Crisis: Synergistic Impacts of Climate Change and Human Activities and Their Feedback Mechanisms*. *Water* 2026, 18(6), 732. <https://doi.org/10.3390/w18060732>
- [34] Food and Agriculture Organization of the United Nations (FAO). *Drought and Agriculture*. <https://www.fao.org/land-water/water/drought/drought-and-agriculture/en/>